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MTH707: Markov chain Monte Carlo

Contents

Contents **1**

Preliminary **3**
Course overview and introduction 3

Markov chain basics **7**
Markov chains basics 7
Target distribution 11
Example: Metropolis-Hastings 12
Stochastic Stability 16

Types of MH Algorithms **21**
Independence MH 21
Random Walk MH 22
Symmetric MH 22
Gradient-based Proposals 22

Bibliography **31**

Preliminary

Course overview and introduction

Lecture 1

This is a graduate level (PhD level) course on Markov chain Monte Carlo (MCMC). Unlike other MCMC courses, this will focus specifically on the theoretical analysis of MCMC samplers; their rates of convergence, and theoretical tools for their analysis. You will require:

- a familiarity with probability theory and a willingness to learn some basic measure theory (on your own)
- a familiarity with Bayesian statistics and a willingness to learn some basic Bayesian modeling strategies (on your own)
- a familiarity with R and a willingness to learn coding algorithms in R

Please be aware that:

> *This is NOT a course where you will learn different algorithms. You might learn different algorithms along the way, but the course focuses on why these algorithms work (theoretically).*

If you're familiar with statistics, you are aware that a critical building block of statistical analysis is *data*. Once give data regarding a certain problem, we use statistical methods to analyze the data. This is *not* what happens in MCMC. Please be aware that:

> *We will NOT be dealing with real data at almost any point in this course.*

MCMC is a computational simulation technique used to obtain samples from complicated distributions. That it, MCMC actually *produces* data from a computer. These data are generated specifically to answer some key mathematics and statistical questions of interest.

One of the main uses of MCMC algorithms is in estimating complicated integrals.

Example 1 (Estimating integrals)**.** Suppose we want to integrate

$$\mu = \int_0^\infty \frac{1}{(1+x)^{2.3}(\log(x+3))^2} dx\,.$$

We can rewrite this as

$$\begin{aligned}\mu &= \int_0^\infty \frac{1}{(1+x)^{2.3}(\log(x+3))^2} dx \\ &= \int_0^\infty \frac{e^x}{(1+x)^{2.3}(\log(x+3))^2} e^{-x} dx \\ &= \mathrm{E}_F\left[\frac{e^x}{(1+x)^{2.3}(\log(x+3))^2}\right],\end{aligned}$$

where F is the $\mathrm{Exp}(1)$ distribution. So if we want to estimate the integral μ, we can generate draws from $\mathrm{Exp}(1)$ and then use these generate samples to estimate μ.

Thus, we would want to obtain a *sample* from the distribution F. That is, we want

$$X_1, X_2, \ldots, X_n \sim F$$

For a general F, this is a non-trivial problem.

- How do we even draw samples from nice and known distributions? Done in MTH210
  - Inversion method (Exponential example)
  - Accept-reject sampling
  - Ratio of uniforms
- How do we draw samples from nice unknown distributions? Done in MTH210
  - Accept-reject sampling
  - Ratio of uniforms
- How do we draw samples from bad unknown distributions? What are bad distributions?
  - Suppose support of the distribution is $\mathbb{R}^{1000}$.
  - F has a very complicated density associated with it.

Below is an example of a complicated distribution from which we want to sample from.

Example 2 (Bayesian one-way random effects model)**.** Suppose we observed real data Y_{ij} for $i = 1, \ldots, k$ and $j = 1, \ldots, m_i$. This, for example, could denote exam scores for the ith student in their jth course.

Each student has an average performance capability, θ_i, so that, we assume

$$Y_{ij} \mid \theta_i, \lambda_e \sim N(\theta_i, \lambda_e^{-1})$$

where λ_e is the inverse variance, also known as the precision. In a Bayesian setting, we assign a *prior* distribution on the known parameters, so that

$$\theta_i \mid \mu, \lambda_\theta \sim N(\mu, \lambda_\theta^{-1})$$
$$\mu \sim N(m_o, s_o^{-1}) \quad \lambda_e \sim \text{Gamma}(a_1, b_1) \quad \lambda_\theta \sim \text{Gamma}(a_2, b_2)$$

Having observed the data, we can update the prior information on $\Theta = (\theta, \mu, \lambda_e, \lambda_\theta)$. This yield the posterior distribution which has density

$$\begin{aligned} q(\Theta|y) &= q(\theta, \mu, \lambda_e, \lambda_\theta \mid y) \\ &= \frac{f(y|\theta, \mu, \lambda_e, \lambda_\theta) f(\theta, \mu, \lambda_e, \lambda_\theta)}{\int_\Theta f(y|\theta, \mu, \lambda_e, \lambda_\theta) f(\theta, \mu, \lambda_e, \lambda_\theta)} \\ &\propto f(y|\theta, \lambda_e) f(\theta|\mu, \lambda_\theta) f(\mu) f(\lambda_\theta) f(\lambda_e)\,. \end{aligned}$$

This final density is proportional to the produce of many densities and has a fairly convoluted resulting distribution. However, in order to estimate Θ, we must understand what $q(\Theta|y)$ is. For example, we may be interested in the quantity $\mathrm{E}[\Theta|y]$.

We will learn how to use MCMC algorithms to answer estimation questions like these. □

<u>*Markov chain*</u> <u>*Monte Carlo*</u> algorithms have two main components.

- <u>*Markov chain*</u>: Given *any* proper distribution F, we will learn how to construct Markov chains in order to obtain a sample from F:

$$X_1, \ldots, X_n \approx F$$

- <u>*Monte Carlo*</u>: Having obtained the above *sample*, how can we estimate quantities pertaining to F? For example, the mean of F, $\mu = \mathrm{E}_F[X]$ could be estimated with

$$\hat{\mu}_n = \frac{1}{n} \sum_{t=1}^{n} X_t$$

  but is this estimate any good, given that the original samples are not iid?

Below is a link to a demo MCMC algorithm that draws samples from a standard bivariate normal distribution $N_2(0, I_2)$. `https://chi-feng.github.io/mcmc-demo/app.html?algorithm=RandomWalkMH&target=standard`

Since these are not iid samples, I can have the following questions:

1. How do we know that the samples are actually from $N_2(0, I_2)$?
2. If my first point is somehow drawn from $N_2(0, I_2)$, are all points $N_2(0, I_2)$?
3. If my first point is not $N_2(0, I_2)$, then what happens to all the points? Are they eventually from $N_2(0, I_2)$?
4. Does a strong law of large numbers hold?
5. Does a Markov chain version of the central limit theorem hold?
6. What is the variance in the Markov chain CLT and how do we estimate it?
7. How do we assess the quality of the samples?
8. When do we stop sampling the Markov chain?

By the end of the course, we should be able to answer all of these questions.

Markov chain basics

In this chapter we will study some of the basics of Markov chains on general state spaces. It will be useful to remember that we are NOT in finite spaces.

Markov chains basics

Let $(\mathcal{X}, \mathcal{B}(\mathcal{X}))$ be a measurable space. We will usually work with $\mathcal{X} \subseteq \mathbb{R}^d$ and the Borel σ-fields, and this will be the underlying space assumed, unless mentioned otherwise. We will also almost exclusively be discussing *discrete-time* continuous (general-state) space Markov chains.

Definition 1 (Time-homogeneous Markov chain)**.** The $\mathcal{X}$-valued sequence of random variables $\Phi = \{X_0, X_1, X_2, \dots\}$ is a time-homogeneous Markov chain if for all $A \in \mathcal{B}(\mathcal{X})$ and for all n

$$\Pr(X_{n+1} \in A \mid X_n, \dots, X_1, X_0) = \Pr(X_{n+1} \in A \mid X_n)\,.$$

There are two properties embedded within this:

- The Markov property: that the distribution of X_{n+1} given $X_n, X_{n-1}, \dots, X_0$ is the same as the distribution of X_{n+1} given X_n.
- Homogeneous transition: the distribution of X_{n+1} given X_n is the same for all n.

The key operating object for a discrete-time general state-space Markov chain is its *Markov transition kernel*.

Definition 2 (Markov transition kernel)**.** A Markov transition kernel is a map $P : \mathcal{X} \times \mathcal{B}(\mathcal{X}) \to [0, 1]$ such that

1. for all $A \in \mathcal{B}(\mathcal{X})$, $P(., A)$ is a measurable function on $\mathcal{X}$.
2. for all $x \in \mathcal{X}$, $P(x, .)$ is a probability measure on $\mathcal{B}(\mathcal{X})$.

Informally, this is just like a conditional probability. For $x \in \mathcal{X}$ and $A \in \mathcal{B}(\mathcal{X})$

$$P(x, A) := \Pr(X_1 \in A \mid X_0 = x)\,.$$

Remark 1. When $\mathcal{X}$ is discrete, $\mathcal{B}(\mathcal{X})$ is the set of all subsets of $\mathcal{X}$ and P is a matrix of transition probabilities having elements $P(x, y)$ s.t. $P(x, y) \geq 0$ and $\sum_y P(x, y) = 1$. We will almost *never* discuss this case.

Definition 3 (Markov transition density)**.** Let $P(x, \cdot)$ be absolutely continuous with respect to a measure μ. Denote $k : \mathcal{X} \times \mathcal{X} :\to [0, \infty)$ as the Markov transition density defined as

$$k(x, y)\mu(dy) = P(x, dy)\,.$$

Example 3. Let $\mathcal{X} = (0, 1)$. Consider the following Markov chain, given an X_0.

(i) Draw $U \sim U(0, 1)$.

(ii) If $U \leq 1/2$, then $X_{n+1} \sim U(0, X_n)$

(iii) and if $U > 1/2$, then $X_{n+1} \sim U(X_n, 1)$.

Then for $X_n = x$

$$\begin{aligned}
&P(x, A) \\
&= \Pr(X_{n+1} \in A | X_n = x) \\
&= \Pr\left(X_{n+1} \in A, U \leq \frac{1}{2} | X_n = x\right) + \Pr\left(X_{n+1} \in A, U > \frac{1}{2} | X_n = x\right) \\
&= \Pr\left(X_{n+1} \in A | U \leq \frac{1}{2}, X_n = x\right) \Pr\left(U \leq \frac{1}{2}\right) \\
&\qquad + \Pr\left(X_{n+1} \in A | U > \frac{1}{2}, X_n = x\right) \Pr\left(U < \frac{1}{2}\right) \\
&= \int_A \frac{1}{2}\frac{1}{x} I_y((0, x)) + \frac{1}{2}\frac{1}{1-x} I_y((x, 1))\, dy\,.
\end{aligned}$$

Here, $P(x, \cdot)$ is absolutely continuous with respect to the Lebesgue measure, we get that the Markov transition density is

$$k(x, y) = \frac{1}{2}\frac{1}{x} I_y((0, x)) + \frac{1}{2}\frac{1}{1-x} I_y((x, 1))\,.$$

□

Let F be a probability measure on $\mathcal{B}(\mathcal{X})$. Define shorthand notation

$$FP(A) := \int_{\mathcal{X}} P(x, A) F(dx)\,.$$

If $X_n \sim F$ and $X_{n+1} | (X_n = x) \sim P(x, \cdot)$ then FP is the marginal distribution of X_{n+1}.

Similarly, if $X_1 \mid X_0 \sim P(x, \cdot)$, then we can define the marginal distribution of $X_2 \mid X_0$ as

$$P^2(x, A) = \Pr(X_2 \in A \mid X_0 = x) = \int_{\mathcal{X}} P(x, dx_1) P(x_1, A)\,.$$

Let's take a minute to be careful about this. The above is the *marginal* distribution of $X_2|X_0$. Even though we are still conditioning on X_0, it is still called the *marginal* distribution since we are integrating over all possible values of X_1.

In this way, we get the n-step Markov transition kernel P^n, which describes the transition after n steps given the starting step.

$$P^n(x, A) = \Pr(X_n \in A \mid X_0 = x) = \int_{\mathcal{X}} P(x, dx_1) P(x_1, dx_2) \ldots, P(x_{n-1}, A)\,. \tag{1}$$

(Chapman-Kolmogorov Equation) In general, for $0 \leq m \leq n \in \mathbb{N}$, $P^n = P^{n-m} P^m$ and thus,

$$P^n(x, A) = \int_{\mathcal{X}} P^m(x, dy) P^{n-m}(y, A)\,.$$

Given a Markov chain transition kernel P, and a starting value $X_0 = x$, we generate the Markov chain $\Phi = \{X_0, X_1, X_2, \ldots\}$

Definition 4. Let F be a distribution such that $X_0 \sim F$. The distribution F is the *invariant* or *stationary* distribution if $FP = F$. That is, if for all $A \in \mathcal{B}(\mathcal{X})$

$$FP(A) = \int_{\mathcal{X}} P(x, A) F(dx) = F(A)\,.$$

This means that if we start from F, and use P to get another sample, then the next sample is also from F.

Theorem 1. *If F is the invariant distribution of P, then for all $n \geq 1$, $FP^n = F$.*

Proof. First, recall that $P^n(\cdot, A)$ is a non-negative measurable function bounded by 1, and hence integrable with respect to probability measures. Since F is invariant for P means that $FP = F$, and notation

$$FP^n(A) = \int_{\mathcal{X}} F(dx) P^n(x, A)\,.$$

Using the Chapman-Kolmogorov equation for $m = 1$

$$= \int_x F(dx) \int_y P(x, dy) P^{n-1}(y, A)$$

$$
\begin{aligned}
&= \int_x \int_y F(dx)P(x,dy)P^{n-1}(y,A) \\
&= \int_y \left(\int_x F(dx)P(x,dy) \right) P^{n-1}(y,A) \qquad \text{By Fubini's theorem} \\
&= \int_{\mathcal{X}} F(dy)P^{n-1}(y,A) \\
&= \vdots \\
&= \int_{\mathcal{X}} F(dy)P(y,A) \\
&= F(A)\,.
\end{aligned}
$$

Note that, the change of integral is assured by Fubini's theorem since $P^{n-1}(y,A)$ is a non-negative measurable and integrable function (since $FP(A) < \infty$ for all A). $\square$

Our goal in this course, however, is not to learn all about Markov chains. What we study about Markov chains will be in the context of MCMC. Recall that in MCMC, our goal is to sample from the distribution F. If we start from F, and construct P so that F is invariant, then each new sample will be from F. This is exactly what we want, so that is great! Now there are still two concerns:

1. we can't really start from F in most problems, so what happens then?
2. and how do we construct P?

We will first answer the second question and learn ways of constructing such P, and then later in the course get to question 1.

A key feature we will use in constructing P such that F is invariant is that of *detailed balance* or *F-symmetric*.

Definition 5. The kernel P is *F-symmetric* if

$$F(dx)P(x,dy) = F(dy)P(y,dx)\,.$$

This equation is also often called **detailed balance**, and the phenomenon is also called F-reversibility.

Theorem 2. *F-symmetry implies F is invariant for P.*

Proof. Need to show that $FP = F$.

$$
\begin{aligned}
FP(A) &= \int_{\mathcal{X}} F(dx)P(x,A) \\
&= \int_{\mathcal{X}} \int_A F(dx)P(x,dy)
\end{aligned}
$$

$$
\begin{aligned}
&= \int_{\mathcal{X}} \int_A F(dy) P(y, dx) \qquad \text{using } F\text{-reversibility} \\
&= \int_A F(dy) \int_{\mathcal{X}} P(y, dx) \\
&= \int_A F(dy) \\
&= F(A)\,.
\end{aligned}
$$

□

Recall that our goal is to construct an F-invariant transition P. So, if we construct P such that it is F-symmetric (reversible), then we are good to go.

Target distribution

Before we continue to construct Markov chains that have invariant distribution F, let's understand the kind of distributions that are of interest.

Recall that F is a target distribution of interest. Since we are focusing on general state spaces, we make the reasonable assumption that F is a distribution that is absolutely continuous with respect to the Lebesgue measure so that $f(x)$ is the associated density function. That is for $A \in \mathcal{B}(\mathcal{X})$

$$F(A) = \int_A f(x) dx\,.$$

In most practical MCMC problems, the pdf $f(x)$ is such that $f(x) = c\tilde{f}(x)$ where $\tilde{f}(x)$ is known, but the constant c is unknown.

Example 4 (Example 2 continued)**.** Recall the Bayesian model

$$Y_{ij} \mid \theta_i, \lambda_e \sim N(\theta_i, \lambda_e^{-1})$$

where λ_e is precision. Further prior distributions are

$$
\begin{gathered}
\theta_i \mid \mu, \lambda_\theta \sim N(\mu, \lambda_\theta^{-1}) \\
\mu \sim N(m_o, s_o^{-1}) \quad \lambda_e \sim \text{Gamma}(a_1, b_1) \quad \lambda_\theta \sim \text{Gamma}(a_2, b_2)
\end{gathered}
$$

This yields the posterior distribution which has density

$$
\begin{aligned}
f(\Theta|y) &\propto p(y|\theta, \lambda_e) p(\theta|\mu, \lambda_\theta) p(\mu) p(\lambda_\theta) p(\lambda_e) \\
&= \prod_i \left[\prod_j \sqrt{\frac{\lambda_e}{2\pi}} \exp\left\{ -\frac{\lambda_e (y_{ij} - \theta_i)^2}{2} \right\} \right] \left[\sqrt{\frac{\lambda_\theta}{\sqrt{2\pi}}} \exp\left\{ -\frac{\lambda_\theta (\theta_i - \mu)^2}{2} \right\} \right] \\
&\quad \times \left[\sqrt{\frac{s_o}{2\pi}} \exp\left\{ -s_o \frac{(\mu - m_0)^2}{2} \right\} \right] \frac{b_1^{a_1}}{\Gamma(a_1)} \lambda_e^{a_1 - 1} e^{-\lambda_e b_1} \frac{b_2^{a_2}}{\Gamma(a_2)} \lambda_\theta^{a_2 - 1} e^{-\lambda_\theta b_2}\,,
\end{aligned}
$$

You can see that the final posterior distribution is $k+3$ dimensional and the proportionality constant is impossible to obtain. □

Such target distributions with *intractable* density functions are a regular feature in Bayesian methods and other applications, and it is here where MCMC finds the most use. As we go along in the course, please keep in mind the complexity of F arising from such problems.

Example: Metropolis-Hastings

Before we start reading more theoretical properties of Markov chains, let's get introduced to an important MCMC algorithm - the Metropolis-Hastings algorithm. This will help build intuition and give some relevance to the kinds of questions we want to ask and answer. The Metropolis-Hastings (MH) algorithm was proposed by Nicholas Metropolis, Mashall Rosenbluth, Arianna Rosenbluth, Augusta Teller, and Edward Teller in a 1953 paper at the Los Alamos Lab in New Mexico, USA. This was followed up with a paper in 1970 by Wilfred Hastings with a generalization. The consequent algorithm is called the "Metropolis-Hastings algorithm" (although it is debated whether more credit should be given to the other authors in the first paper).

Let Q be a Markov kernel so that $Q(x, \cdot)$ is a distribution with transition density $q(x, y)$. That is,

$$Q(x, dy) = q(x, y)dy\,.$$

Although we call this a 'kernel', think of this as a distribution conditioned on x. The distribution $Q(x, \cdot)$ is such that it is easy to draw samples from $Q(x, \cdot)$. For example, we will often set $Q(x, \cdot) = N(x, h)$, where the corresponding density is

$$q(x, y) = \frac{1}{\sqrt{2\pi}} e^{-(y-x)^2/(2h)}\,.$$

The Metropolis-Hastings MCMC algorithm proposed values from $Q(x, \cdot)$ and then accepts or rejects them in a way that keeps F invariant.

Metropolis-Hastings algorithm

Let $X_n = x$. To obtain X_{n+1}:

1. $Y = y \sim Q(x, .)$
2. Set $X_{n+1} = y$ with probability
$$\alpha_{\text{MH}} = \min\left\{1, \frac{f(y)\,q(y, x)}{f(x)\,q(x, y)}\right\}$$
3. Set $X_{n+1} = x$ with probability $1 - \alpha_{\text{MH}}$.

The above must be implemented on a computer. The act of doing something with a probability, can be done by using $U(0,1)$ draws. Thus, practically, the MH algorithm becomes:

Metropolis-Hastings algorithm on a computer

Let $X_n = x$. To obtain X_{n+1}:

1. $Y = y \sim Q(x, .)$ and independently $U \sim \text{Unif}(0,1)$
2. If
$$U \leq \min\left\{1, \frac{f(y)\,q(y,x)}{f(x)\,q(x,y)}\right\}$$
then set $X_{n+1} = y$
3. Else set $X_{n+1} = x$

The ration $r(x,y)$ is called the Hasting's ratio where

$$r(x,y) = \frac{f(y)q(y,x)}{f(x)q(x,y)}$$

and $\alpha_{\text{MH}}(x,y) = \min\{1, r(x,y)\}$ is called the *Hasting's acceptance probability*. Later, we will study other acceptance probabilities as well. Notice that if $f(x) = c\tilde{f}(x)$, then

$$r(x,y) = \frac{\tilde{f}(y)q(y,x)}{\tilde{f}(x)q(x,y)}\,,$$

and thus, not knowing the constant c does not create any problems in implementing the algorithm.

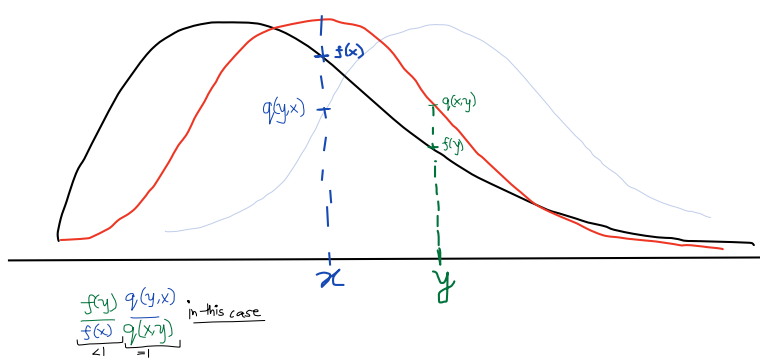

Figure 1: Metropolis-Hastings algorithm: some intuition. The target density is in black, and the proposal density is in red ($q(x,y)$). The blue is the reversed proposal, $q(y,x)$. Here, the point proposed (y) is in a lower probability region, so there are chances to reject.

Note, for a given x, $\delta_x(A) = 1$ only if $x \in A$. This is the Dirac measure on the set A.

Theorem 3. *The MH algorithm defines the following Markov transition kernel:*

$$P(x, A) = \int_A Q(x, dy)\, \alpha(x, y) + \delta_x(A) \int_{\mathcal{X}} [1 - \alpha(x, u)]\; Q(x, du)$$

Proof. By definition, $P(x, A)$ is

$$\Pr(X_{n+1} \in A \mid X_n) = \underbrace{\Pr(X_{n+1} \in A, U \leq \alpha(X_n, Y) \mid X_n)}_{I} + \underbrace{\Pr(X_{n+1} \in A, U > \alpha(X_n, Y) \mid X_n)}_{II}\,.$$

We will look at I and II separately. First,

$$\begin{aligned}
I &= \Pr(X_{n+1} \in A, U \leq \alpha(X_n, Y) \mid X_n = x) \\
&= \mathrm{E}\Big[\mathbb{I}(Y \in A, U \leq \alpha(X_n, Y)) \mid X_n = x\Big] \\
&= \mathrm{E}\Big[\mathrm{E}\Big[\mathbb{I}(Y \in A)\mathbb{I}(U \leq \alpha(X_n, Y)) \mid Y\Big] \mid X_n = x\Big] \\
&= \mathrm{E}\Big[\mathbb{I}(Y \in A)\mathrm{E}\Big[\mathbb{I}\left(U \leq \alpha(X_n, Y)\right) \mid Y\Big] \mid X_n = x\Big] \\
&= \mathrm{E}\left[\mathbb{I}(Y \in A)\alpha(X_n, Y) \mid X_n = x\right] \\
&= \int_A \alpha(x, y) Q(x, dy)\,.
\end{aligned}$$

$$\begin{aligned}
II &= P(X_{n+1} \in A, U > \alpha(X_n, Y) \mid X_n = x) \\
&= \mathrm{E}\Big[\mathrm{E}\Big(\mathbb{I}(X_n \in A)\,\mathbb{I}(U > \alpha(X_n, Y)) \mid Y\Big) \mid X_n = x\Big] \\
&= \mathrm{E}\Big[\mathbb{I}(X_n \in A)\,\mathrm{E}\Big(\mathbb{I}(U > \alpha(X_n, Y)) \mid Y\Big) \mid X_n = x\Big] \\
&= \mathbb{I}(X_n \in A)\mathrm{E}\Big[(1 - \alpha(X_n, Y)) \mid X_n = x\Big] \\
&= \delta_x(A) \int_{\mathcal{X}} (1 - \alpha(x, u)) Q(x, du)\,.
\end{aligned}$$

□

Remark 2. Assuming all distributions are absolutely continuous with respect to the Lebesgue measure, the corresponding Markov transition density is

$$k(x, y) = q(x, y)\alpha(x, y) + \delta_x(y) \int (1 - \alpha(x, u)) q(x, u) du\,.$$

Note that the reference measure w.r.t which we get the above density is a mixture measure.

Now that we know the form of the kernel and we know how to implement the algorithm, we would first like to see whether it is F-invariant. We know one method to check this: F-symmetric implies F-invariance by Theorem 2.

Theorem 4. *The Metropolis-Hastings kernel is F-symmetric.*

Proof. To show F-symmetric, we need to show that

$$F(dx)P(x, dy) = F(dy)P(y, dx)$$

This is trivially true if $x = y$. So it suffices to consider the case when $x \neq y$, which is the case when the MH algorithm moves. That is

$$\begin{aligned}
F(dx)P(x, dy) &= f(x)\, q(x, y)\, \alpha(x, y)\, dx\, dy \\
&= f(x)\, q(x, y) \min\left\{1, \frac{f(y)\, q(y, x)}{f(x)\, q(x, y)}\right\} dx\, dy \\
&= \min\left\{f(x)\, q(x, y), f(y)\, q(y, x)\right\} dx\, dy \\
&= f(y)\, q(y, x) \min\left\{1, \frac{f(x) q(x, y)}{f(y) q(y, x)}\right\} dx\, dy \\
&= F(dy)P(y, dx)
\end{aligned}$$

□

Example 5 (χ^2 target distribution)**.** Consider drawing from a χ^2-distribution. Of course, this is not a complicated distribution, we already know how to draw from it. But for the sake of demonstration:

$$f(x) = \frac{1}{2^{k/2}\Gamma(k/2)} x^{k/2-1} e^{-x/2}\, I(x > 0)\,.$$

To implement a MH algorithm, we need to choose a proposal distribution. Let $Q(x, \cdot) = N(x, h)$, where h is the variance. The density is

$$q(x, y) = \frac{1}{\sqrt{2\pi h}} \exp\left\{-\frac{(y-x)^2}{2h}\right\} = \frac{1}{\sqrt{2\pi h}} \exp\left\{-\frac{(x-y)^2}{2h}\right\} = q(y, x)\,.$$

Here, the acceptance ratio is

$$\alpha(x, y) = \min\left\{1,\ \frac{f(y)}{f(x)} \frac{q(y, x)}{q(x, y)}\right\} = \min\left\{1,\ \frac{f(y)}{f(x)}\right\} = \min\left\{1,\ \frac{y^{k/2-1} e^{-y/2}}{x^{k/2-1} e^{-x/2}}\right\}$$

We now have all the tools to implement the MH algorithm. Note that we don't have to worry about $f(x) = 0$ in this case as long as we start from within our support of the target distribution. This is because, x will only be a new accepted value if $f(x) > 0$.

Types of MH Proposals: We will spend a lot of time studying the different types of MH proposals. For now, though, we require the tools to assess whether a particular proposal will be effective or not.

Stochastic Stability

We now know how to construct a Metropolis-Hastings algorithm that keeps F-invariant. This implies that if the Markov chain starts from F, then it will continue sampling from F. However, what if we can't sample from F? What qualities of the Markov chain will be required to obtain representative samples from F?

Effectively, this will come down to constructing Markov chains that "explore the space $\mathcal{X}$". We will require some additional structure to the Markov chain Φ.

Definition 6 (F-communicating)**.** A set $A \in \mathcal{B}(\mathcal{X})$ is F-communicating, if for all $B \subset A$ such that $F(B) > 0$ and $B \in \mathcal{B}(\mathcal{X})$ and for all $x \in A$ there exists n such that

$$P^n(x, B) > 0\,.$$

Intuitively, the chain will comeback to anywhere in A eventually. This is a weak property, since it does not ask the chain to move out of A.

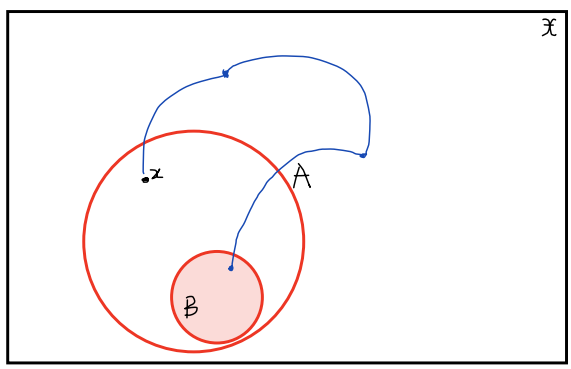

Figure 2: Pictorial depiction of F-communicating. Here we can go from x to B in 3 steps.

Definition 7 (F-irreducible)**.** P is F-irreducible if for all $x \in \mathcal{X}$ and $A \in \mathcal{B}(\mathcal{X})$ such that $F(A) > 0$ there exists n such that

$$P^n(x, A) > 0\,.$$

Otherwise P is reducible.

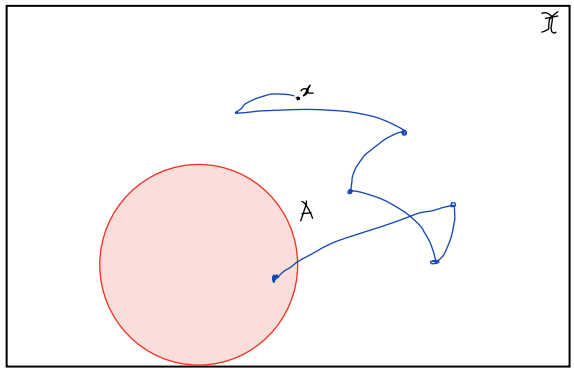

Figure 3: Pictorial depiction of F-irreducible. Here we can go from x to A in 6 steps.

Example 6 (Broken support)**.** Consider the following density:

$$f(x) = \frac{1}{4} I(0 < x < 2) + \frac{1}{4} I(3 < x < 5)\,.$$

and let F be the associated distribution

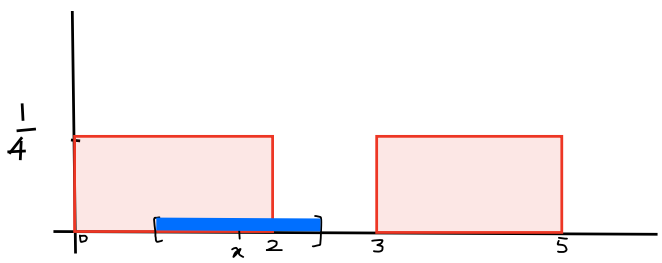

Figure 4: Given x, we cannot jump support in this case.

However, consider a MH kernel with proposal $Q(x, \cdot) = U(x-1, x+1)$ and target F. Then the chain is not F-irreducible since if $X_0 = x \in (0, 2)$ the set $A = (3, 5)$ can never be reached. The kernel P would be F-irreducible if kernel made jumps in $(x - 1.2, x + 1.2)$ for example.

Then the resulting P is reducible.

Remark 3. Positivity condition: A trivial way of getting F-irreducibility is to have the positivity condition - construct P so that $P(x, A) > 0$ for all A and x! We will keep using this often.

Example 7 (Limited jumps)**.** Consider the density

$$f(x) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{x^2}{2}\right\}\,.$$

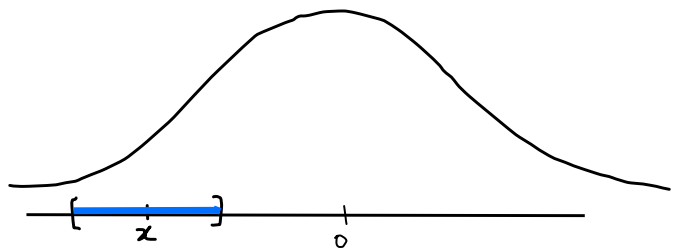

Figure 5: Given x, we can explore the whole support.

Let F be the associated distribution. Consider the MH proposal

$$Q(x, A) = U(x - 1, x + 1)\,.$$

It is clear that P is irreducible since it is possible to go from anywhere to anywhere in some amount of steps. But showing this mathematically is hard.

Theorem 5 (Geyer (1998), Theorem 4.1)**.** *Suppose*

a. The state space $\mathcal{X}$ is a separable metric space.

b. Every non-empty open set $A \in \mathcal{B}(\mathcal{X})$ is such that $F(A) > 0$

c. Every point has a F-communicating neighborhood

then the chain is F-irreducible.

Proof. The proof is complicated, but the intuition can be seen from Example 7. First, $\mathcal{X}$ is a separable metric space - there exists a countable collection of open sets U such that even open subset of $\mathcal{X}$ is a union of sets from U.

Condition (b) for our purpose means that the distribution F has full support $\mathcal{X}$, and (c) means that given any point x, we can always go to any subset of its neighborhood.

These three points eventually allow movement from anywhere to anywhere in finite steps. □

Theorem 6. *For the MH algorithm, if the proposal kernel $Q(x, \cdot)$ with density $q(x, y)$ is such that $q(x, y) > 0$ for all $x, y \in \mathcal{X}$, then P is F-irreducible.*

Proof. Homework (will be assigned). □

If $q(x, y) > 0$ for all x, y then $P(x, A) > 0$ for all $F(A) > 0$, thus P is F-irreducible. In cases when this is not true, Geyer's theorem can be used in most cases to establish irreducibility.

Thus, we end this section with some insight on how to choose a "good" proposal distribution for the MH algorithm. What types of proposals satisfy this assumption? What are examples of good proposal distributions. We will discuss this in the next chapter.

Definition 8 (Aperiodic)**.** For $d \geq 2$, consider disjoint sets $A_1, \ldots, A_d$ such that for $N = (A_1 \cup \cdots \cup A_d)^c$ then $F(N) = 0$. Further, let the sets satisfy $F(A_i) > 0$ and $P(x, A_{i+1}) = 1$ for all $x \in A_i$, $i \leq i \leq d-1$ and $P(x, A_1) = 1$ for all $x \in A_d$. If such sets do not exist, then P is aperiodic.

Otherwise, P is periodic with period d.

Aperiodicity ensures that the chain does not get stuck in a loop that is deterministic upto sets.

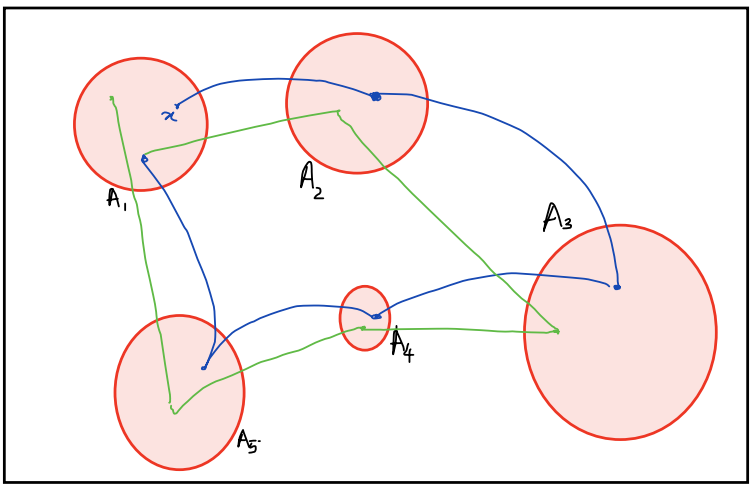

Figure 6: The Markov chain jumps from one set to the other deterministically. The above is a periodic Markov chain.

We will come back to these concepts again when we study convergence of Markov kernels. But these two conditions together, ensure that the Markov chain "explores the state-sapce $\mathcal{X}$" sufficiently well.

Types of MH Algorithms

Recall the MH algorithm where we choose a proposal kernel $Q(x, \cdot)$.

Metropolis-Hastings algorithm

Let $X_n = x$. To obtain X_{n+1}:

1. $Y \sim Q(x, .)$ and independently $U \sim \text{Unif}(0, 1)$
2. If
$$U \leq \alpha(x, y) = \min\left\{1, \frac{f(y)\, q(y, x)}{f(x)\, q(x, y)}\right\} = \min\{1, r(x, y)\}$$
then set $X_{n+1} = y$
3. Else set $X_{n+1} = x$

Clearly, the performance of the MH sampler rests critically on the choice of proposal kernel $Q(x, c)$. We first discuss some different choices of kernels available to us. Remember that for all reasonable proposal distributions, the MH algorithms keeps F-invariant. So then we would like to choose $Q(x, \cdot)$ so that it potentially makes large jumps and is able to sample from regions of high probability fair well.

Independence MH

Here $Q(x, \cdot) = Q(\cdot)$. That is, the proposal distribution does not depend on the current value. Then, $q(x, y) = q(y)$, so the MH-ratio is:

$$r(x, y) = \frac{f(y)\, q(y, x)}{f(x)\, q(x, y)} = \frac{f(y)}{f(x)} \frac{q(x)}{q(y)} = \frac{w(y)}{w(x)}\,,$$

where $w(y) = f(y)/q(y)$ is a weight function. Later, we will also see that bounded weight functions are important here. Naturally, for the resulting Markov kernel to be reversible, the support of Q must be larger than the support of F.

Since our proposals do not "move" with the state, the choice of Q becomes critically important and chosen so that all parts of the support of F can we well-explored. Thus,

independence MH, if it works well, can work *very* well. But if it doesn't work well, then it can be horrible.

Note that, although the proposal distribution doesn't depend on x, the resulting process is still a Markov chain.

Random Walk MH

Suppose $Z \sim G(\cdot)$ so that G is not dependent on the current step X_n. Set,

$$Y = X_n + Z\,.$$

Then, $q(x, y) = g(y - x)$ and

$$r(x, y) = \frac{f(y)\,q(y, x)}{f(x)\,q(x, y)} = \frac{f(y)}{f(x)}\frac{g(x - y)}{g(y - x)}\,.$$

Often, G is either $N(0, h)$ or $U(-h, h)$. But we can use $G = t_d$ proposal distribution.

Symmetric MH

Here, the proposal distribution is such that it exhibits a density q so that $q(x, y) = q(y, x)$. Then, just like in the example discussed before

$$r(x, y) = \frac{f(y)\,q(y, x)}{f(x)\,q(x, y)} = \frac{f(y)}{f(x)}\,.$$

The symmetric MH is the most common proposal, since it makes evaluating the ratio significantly easier.

The two most common proposals are $N(x, h)$ and $U(x - h, x + h)$ for some chosen h. Most Symmetric proposals are also random-walk proposals, but that need not always be the case.

For when the target is d-dimensional, symmetric random -walk proposal: $Q(x, \cdot) = N_d(x, \Omega)$ where Ω is chosen by the user, and is typically a diagonal matrix. In this case:

$$q(x, y) \propto e^{(y-x)^T \Omega^{-1} (y-x)/2}\,,$$

and for a proposed y, the whole vector is either accepted or rejected in one go.

Gradient-based Proposals

So far, the proposals we have discussed only use $X_n = x$ as a "mean parameter", but do not use information about F in the proposal. Naturally, if we are effectively able to use information about the target in the proposal, then we might be able to get MH algorithms that are able to make larger jumps and move more freely in the space.

This is often done via gradients: $\nabla \log f(x)$.

Metropolis-adjusted Langevin Algorithm (MALA)

The symmetric MH with $N(x, h)$ proposes values without utilizing any information from the target. If we can utilize some information from the target, that may help us acquire better proposal distributions. We shift the mean of the proposal distribution towards a higher probability area under F:

$$Q(x, \cdot) = N\left(x + \frac{h}{2}\nabla \log f(x), h\right)$$

Here the mean gets moved in the direction of $\nabla \log f(x)$, moving the proposal closer an area of higher probability.

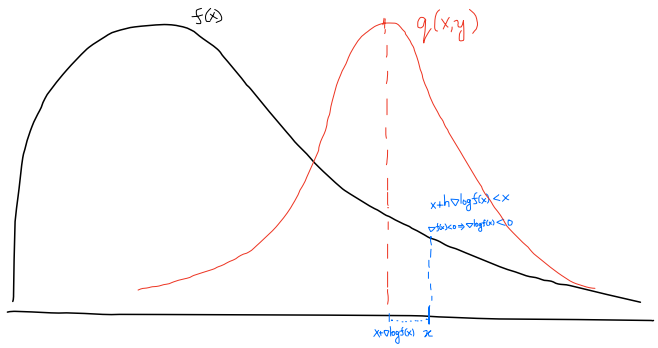

Figure 7: MALA proposal moves the proposal towards a higher target mass area.

Why "Langevin"?: The derivation of this method came from stochastic differential equations. Consider a continuous time process:

$$dX_t = \frac{1}{2}\nabla \log f(X_t)dt + dW_t$$

where W is the standard Brownian motion. This is the "Langevin diffusion" equation and the stationary distribution of this diffusion (under certain conditions) is F. The idea was to run this diffusion to sample from F! However, this is not possible, since we can't run a continuous time process on a computer.

Solution: Euler-Murayama Discretization!

Let us discretize this process by time units of $dt = h$. Then

$$dX_t = X_{t+h} - X_t$$

and

$$dW_t = W_{t+h} - W_t \sim N(0, h)\,,$$

where the increments are independent. So, for $Z_t \overset{\text{iid}}{\sim} N(0, h)$ we get

$$X_{t+h} = X_t + \frac{h}{2}\nabla \log f(X_t) + \sqrt{h}Z_t$$

However, due to the discretization, the Langevin dyamics are not simulated exactly, and the above discretized process does not converge to F by an approximation of F. The error in this approximation is worse when h is large. To correct for this approximation error, the next discretized state is used as the proposal distribution!

$$Y = X_t + \frac{h}{2}\nabla \log f(X_t) + \sqrt{h}Z_t \sim N\left(X_t + \frac{h}{2}\nabla \log f(X_t), h\right) .$$

Example 8 (Bayesian logistic regression)**.** Suppose we observe binary responses $Y_1, Y_2, \ldots, Y_n$ and a popular classification model is logistic regression. When the ith covariate is $(x_{i1}, \ldots, x_{ip})^T$, then for $\beta \in \mathbb{R}^p$ the logistic likelihood assumes the model

$$Y_i \overset{\text{iid}}{\sim} \text{Bern}\left(\frac{e^{x_i^T\beta}}{1+e^{x_i^T\beta}}\right) .$$

In other words, the probability that any response takes the value 1 is

$$\Pr(Y_i = 1) = \frac{e^{x_i^T\beta}}{1+e^{x_i^T\beta}} =: p_i \, .$$

In a Bayesian setup, we assume a prior on β and a common one is:

$$\beta \sim N_p\left(0, \sigma^2 \mathbb{I}_p\right)$$

The posterior distribution is then Our goal is to obtain the MLE of β. As usual, first we write down the log-likelihood.

$$\begin{aligned} f(\beta|y) &= p(y_1, y_2, \ldots, y_n|\beta)p(\beta) \\ &= \left(\prod_{i=1}^n (p_i)^{y_i}\,(1-p_i)^{1-y_i}\right) e^{-\beta^\top\beta/(2\sigma^2)} \, . \end{aligned}$$

In order to implement the MALA algorithm we need the log of the posterior

$$\begin{aligned} \log f(\beta|y) &= \sum_{i=1}^n y_i \log p_i + \sum_{i=1}^n (1-y_i)\log(1-p_i) - \frac{\beta^\top\beta}{2} \\ &= -\sum_{i=1}^n \log\left(1+\exp(x_i^T\beta)\right) + \sum_{i=1}^n y_i x_i^T\beta - \frac{\beta^\top\beta}{2} \, . \end{aligned}$$

Taking gradient,

$$\begin{aligned} \nabla \log f(\beta|y) &= \sum_{i=1}^n x_i y_i - \sum_{i=1}^n \frac{x_i e^{x_i^T\beta}}{1+e^{x_i^T\beta}} - \beta \\ &= \sum_{i=1}^n x_i \left[y_i - \frac{e^{x_i^T\beta}}{1+e^{x_i^T\beta}}\right] - \beta \end{aligned}$$

$$= \sum_{i=1}^{n} x_i \left[y_i - \frac{1}{1+e^{-x_i^T \beta}} \right] - \beta \,.$$

Now the MALA algorithm can be implemented with the proposal for each component independently, which gives

$$Q(\beta_t, \cdot) = N_p \left(\beta_t + \frac{h}{2} \left(\sum_{i=1}^{n} x_i \left[y_i - \frac{1}{1+e^{-x_i^T \beta_t}} \right] - \beta_t \right), h\mathbb{I}_p \right) \,.$$

□

Some challenges with implementing the MALA algorithm are:

- The target density must be differentiable
- The gradient should be available in closed form
- Since every run requires two calls to the gradient, this is more expensive than random-walk Metropolis.

Example 9 (Super-Gaussian distribution)**.** Consider

$$f(x) \propto e^{-x^4} \,, \text{ for } x \in \mathbb{R} \,.$$

Now, for this,

$$\nabla \log f(x) = -4x^3 \,.$$

Notice that, if $x = 2$, then for $h = 1$

$$Y \sim N\left(x + \frac{h}{2}(-4(8)), h \right) = N\left(-14, 1\right) \,.$$

This proposal is too far away and will lead to larger rejections! This makes MALA challenging to implement here. □

The above example is one where the gradient at a point is much larger than the point itself. That is

$$\frac{|\nabla \log f(x)|}{|x|}$$

is large. We will come back to this when we study rates of convergence of Markov chains.

Barker's Proposal

Noticing the lack of robustness of the MALA algorithm to such noisy gradients, Livingstone and Zanella (2022) proposed the Barker's proposal. The proposal density is

$$q_{\mathrm{B}}(x, y) = \frac{2}{1 + e^{-(y-x)\nabla \log f(x)}} \mu_\sigma(y - x) \,.$$

The idea is that instead of moving the mean of the Normal proposal in the direction of the gradient, let us *skew* the normal proposal in the direction of the gradient.

Below I present the evolution of this sampler. Consider function

$$g(t) = \frac{t}{1+t}\,.$$

Then it is established that the kernel described by the density

$$q_{\text{exact}}(x,y) \propto g\left(\frac{f(y)}{f(x)}\right)\mu_\sigma(y-x)\,,$$

converges to F as $\sigma \to 0$. That is, if we choose σ very small, and run the kernel $Q_{\text{exact}}(x,\cdot)$ corresponding to the above density, that kernel will converge to a distribution very close to F. Naturally, we can never set σ exactly to 0. So, just like MALA, if we now use Q_{exact} as our proposal, we should have a nice proposal. However, we can't do this, because

$$q_{\text{exact}}(x,y) = \frac{g\left(\frac{f(y)}{f(x)}\right)\mu_\sigma(y-x)}{Z(x)}\,,$$

where $Z(x)$ is the appropriate normalizing constant. This makes using Q_{exact} impossible since

$$Z(x) = \int g\left(\frac{f(y)}{f(x)}\right)\mu_\sigma(y-x)dy\,.$$

and this integral is not tractable in general. So, a further approximation is made in that we will employ a first-order Taylor series approximation of $f(y)/f(x)$ around $f(x)$:

$$\begin{aligned}
\frac{f(y)}{f(x)} &= e^{\log f(y) - \log f(x)} \\
&= e^{\log f(x) + (y-x)\nabla \log f(x) - \log f(x)} \\
&= e^{(y-x)\nabla \log f(x)} \\
\Rightarrow g\left(\frac{f(y)}{f(x)}\right) &= \frac{e^{(y-x)\nabla \log f(x)}}{1+e^{(y-x)\nabla \log f(x)}} \\
&= \frac{1}{1+e^{-(y-x)\nabla \log f(x)}}\,.
\end{aligned}$$

Thus, an approximation to q_{exact} is

$$q_{\text{B}}(x,y) = \frac{1}{\tilde{Z}(x)}\frac{1}{1+e^{-(y-x)\nabla \log f(x)}}\mu_\sigma(y-x)\,,$$

where $\tilde{Z}(x)$ is the appropriate normalizing constant. Quite magically, it can be shown that for a symmetric μ_σ density, centered at 0,

$$\tilde{Z}(x) = \frac{1}{2}\,.$$

This, then finally reveals:

$$q_{\mathrm{B}}(x, y) = \frac{2}{1 + e^{-(y-x)\nabla \log f(x)}} \mu_\sigma(y - x)\,.$$

This makes it so that the Barker's proposal density is tractaly available and we can indeed calculate $r(x, y)$ in our algorithm. However, recall that we still need a mechanism to sample from this proposal density. The following algorithm samples from the Barker's proposal

Obtaining draws from the Barker's proposal

Let current point be x

1. Draw $Z = z \sim \mu_\sigma(y - x)$
2. Calculate
$$p(x, z) = \frac{1}{1 + e^{-z\nabla \log f(x)}} \in (0, 1)$$
3. Set $b(x, z) = 1$ with probability $p(x, z)$
and $b(x, z) = -1$ with probability $1 - p(x, z)$
4. Set $Y = x + b(x, z) \times Z$.

Then Y is a draw from $Q_{\mathrm{B}}(x, \cdot)$

Proof. We want to show that $\Pr(Y \in A) = Q_{\mathrm{B}}(x, A)$. Consider

$$\begin{aligned}
\Pr(Y \in A) &= \Pr(Z \in A - x, b(x, Z) = 1) + \Pr(-Z \in A - x, b(x, Z) = -1) \\
&= \Pr(Z \in A - x, b(x, Z) = 1) + \Pr(Z \in A - x, b(x, -Z) = -1) \\
&= \int_{A-x} \frac{1}{1 + e^{-z\nabla \log f(x)}} \mu_\sigma(z) dz + \int_{A-x} \left(\frac{1}{1 + e^{z\nabla \log f(x)}} \right) \mu_\sigma(z) dz \\
&= 2 \int_{A-x} \frac{1}{1 + e^{-z\nabla \log f(x)}} \mu_\sigma(z) dz \\
&= \int_{A} \frac{1}{1 + e^{-(y-x)\nabla \log f(x)}} \mu_\sigma(y - x) dy \\
&= Q_{\mathrm{B}}(x, A)\,.
\end{aligned}$$

□

It is then evident from the proposal that the size of any proposed value is not affected by the gradient, but the direction of the proposed value is affected by the gradient.

Barker's Proposal in p dimensions: In multi-dimensions, the proposal employed is the product form proposal with the partial derivatives employed for each component.

$$q_{\mathrm{mBarker}}(x, y) = \prod_{j=1}^{p} q_{\mathrm{Barker}}(x^{(j)}, y^{(j)})$$

Example 10 (Bayesian Poisson Regression)**.** Suppose we observe count responses $Y_1, Y_2, \ldots, Y_n$ and popular model is the Poisson regression model. When the ith covariate is $(x_{i1}, \ldots, x_{ip})^T$, then for $\beta \in \mathbb{R}^p$ the Poisson likelihood assumes the model

$$Y_i \overset{\text{iid}}{\sim} \text{Poisson}\left(e^{x_i^T \beta}\right) .$$

We may assume prior

$$\beta \sim N_p(0, \sigma^2 \mathbb{I}_p) .$$

For this, we have the posterior density

$$f(\beta|y) \propto \left(\prod_{i=1}^{n} \frac{e^{y_i x_i^T \beta} e^{-e^{x_i^T \beta}}}{y!}\right) e^{-\beta^T \beta/(2\sigma^2)}$$

$$\Rightarrow \log f(\beta|y) = \log \text{constant} - \sum_{i=1}^{n} \log y_i! + \sum_{i=1}^{n} y_i x_i^T \beta - e^{x_i^T \beta} - \frac{\beta^T \beta}{2\sigma^2}$$

$$\Rightarrow \nabla \log f(\beta|y) = \sum_{i=1}^{n} y_i x_i - x_i e^{x_i^T \beta} - \frac{\beta}{\sigma^2}$$

If we were to implement MALA for this algorithm, the gradient can often increase dramatically. Consider the example of when $p = 1$. Then if even one of the x_is is positive, then as $\beta \to \infty$

$$\frac{|\nabla \log f(\beta|y)|}{|\beta|} \to \infty .$$

Thus, the Bayesian Poisson regression model is not suitable for MALA, due to the tails of the posterior being very thin. Instead, we can use Barker's proposals quite effectively.

Hamiltonian Monte Carlo

Consider an object moving on a friction-less surface. At any moment in time, the object has both *potential energy* and *kinetic energy*. The potential energy is determined by the height of the object from the ground, and the kinetic energy is determined by its momentum. What does this have to do with our problem statement of sampling from π? Well, consider function

$$U(q) = -\log(\pi(q)).$$

Then a draw $q \sim \pi$ is thought to have potential energy $U(q)$. This is an assumed setup. Below is a pictorial demonstration of $\pi = N(0, 1)$. Points in the tail have large potential energy and points in areas of high probability are given low potential energy.

Now, let's assume the object is moving on the surface described by $U(q)$, with a momentum p. That is the object of mass m has kinetic energy

$$K(p) = \frac{p^2}{2m} .$$

If left undisturbed, this object moves on the surface while conserving its total energy:

$$H(q,p) = U(q) + K(p)\,.$$

The total energy of the system is the Hamiltonian, $H(q,p)$. The Hamiltonian is also incredibly useful in determining how the object will move in a given system. So, suppose you are given information on U and K for an object, and need to determine where the object will be at a given time, t, then Hamilton's equations help us:

$$\frac{dq}{dt} = \frac{\partial H(q,p)}{\partial p} \qquad \text{and} \qquad \frac{dp}{dt} = -\frac{\partial H(q,p)}{\partial q}\,.$$

The solutions to these system of equations will yield $q(t)$ and $p(t)$, the position and momentum at time t, respectively, and will depend on the initial condition: starting position at $t = 0$ and starting momentum at $t = 0$.

Example 11. Let's see a more concrete example of this. As before, consider $\pi = N(0,1)$ and $m = 1$, so that

$$U(q) = \frac{q^2}{2} \quad \text{and} \quad K(p) = \frac{p^2}{2} \quad \Rightarrow \quad H(q,p) = \frac{q^2}{2} + \frac{p^2}{2}\,.$$

Using Hamilton's equations we obtain

$$\frac{dq}{dt} = \frac{\partial H(q,p)}{\partial p} = p \qquad \frac{dp}{dt} = -\frac{\partial H(q,p)}{\partial q} = -q\,.$$

The solutions to these system of PDEs is available analytically. For constants a, r denoting initial conditions, one can obtain the solutions below:

$$q(t) = r\cos(a+t) \qquad \text{and} \qquad p(t) = -r\sin(a+t)$$

which indicates the position at time t is $(q(t), p(t))$, when the object starts its motion at $(q(0), p(0)) = (r\cos(a), -r\sin(a))$. In the figure below (left), I set $r = 3$ and $a = 1$, so that $q(0) = 1.621$ and $p(0) = -2.52$. This is the point where the particle starts — from the location 1.621 with momentum -2.52. Then I draw the joint path of $(q(t), p(t))$ as time evolves (which just describes a circle here). Since the initial energy is negative, when I progress the particle for $t = 2$ time units, it decreases the location component q. In the left is the corresponding potential energy function, $U(q) = -\log \pi(q)$. □

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